IPv6: network transition

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Credits: this presentation is partly based
on a set of slides made from Ivan Pepelnjak and available here:

http://www.slideshare.net/IOSHints/nat64-and-dns64-in-30-minutes

6PE

- 6PE may offer a nice answer to any NSP whose core network runs MPLS
- Idea
  - Keep the core unchanged
    - (not excluding that it can evolve in the future)
  - Add support fo IPv6 networks at the edge
    - Networks giving IPv6 connectivity to end users
  - Distribute IPv6 routing information in MPLS/BGP, in the same way as we currently do with VPNs

Leclerc

- [RFC 4798] J. De Clercq , D. Ooms, S. Prevost, F. Le
Vasseur
Faucheur, "Connecting IPv6 Islands over IPv4 MPLS Using IPv6 Provider Edge Routers (6PE)," February 2007.

Possible options for IPv6 over MPLS networks

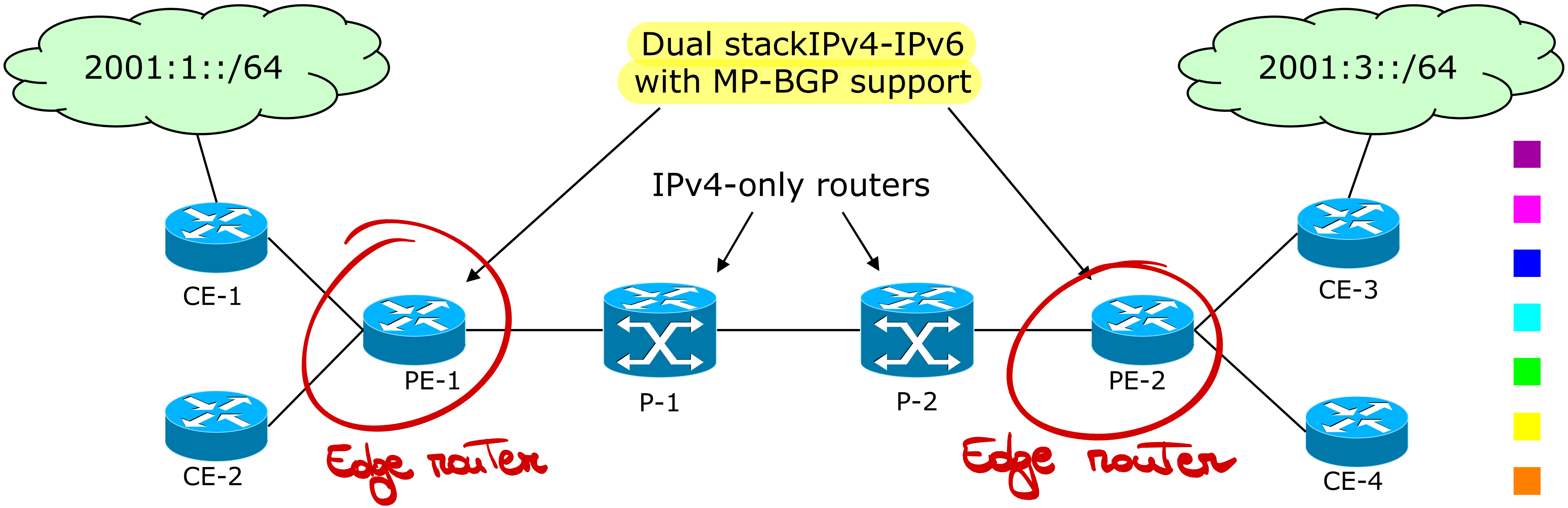
- (1) Native IPv6 over MPLS
  - IPv6 and IPv4 traffic is treated identically by the core routers
  - Core Infrastructure requires full Control Plane upgrade to IPv6
    - IPv6 Routing in core
    - IPv6 Label Distribution Protocol in core
    - Dual Control Plane management if IPv4 and IPv6 services
- (2) IPv6 over Circuit Transport over MPLS
  - L2 frames, e.g. Ethernet frames or ATM cells, are encapsulated into MPLS frames and transported over the network
  - No changes are needed to Provider routers
  - Scalability problems arise in heavily (L2 tunnel) mesh topologies
- (3) 6PE
  - Similar to MPLS VPNs in terms of technical implementation and complexity

6PE: applicability

- 6PE is a good choice for NSPs that have MPLS core network and supports MPLS VPN (or other) services
- IPv6 services are requested by a limited number of customers
  - If the number of IPv6 customers increases so that the NSP requires most of the access routers to become 6PE, the NSP should consider to upgrade to whole network
- The ISP wants to avoid either to fully upgrade the core network or to deploy IPv6-over-IPv4 tunnels

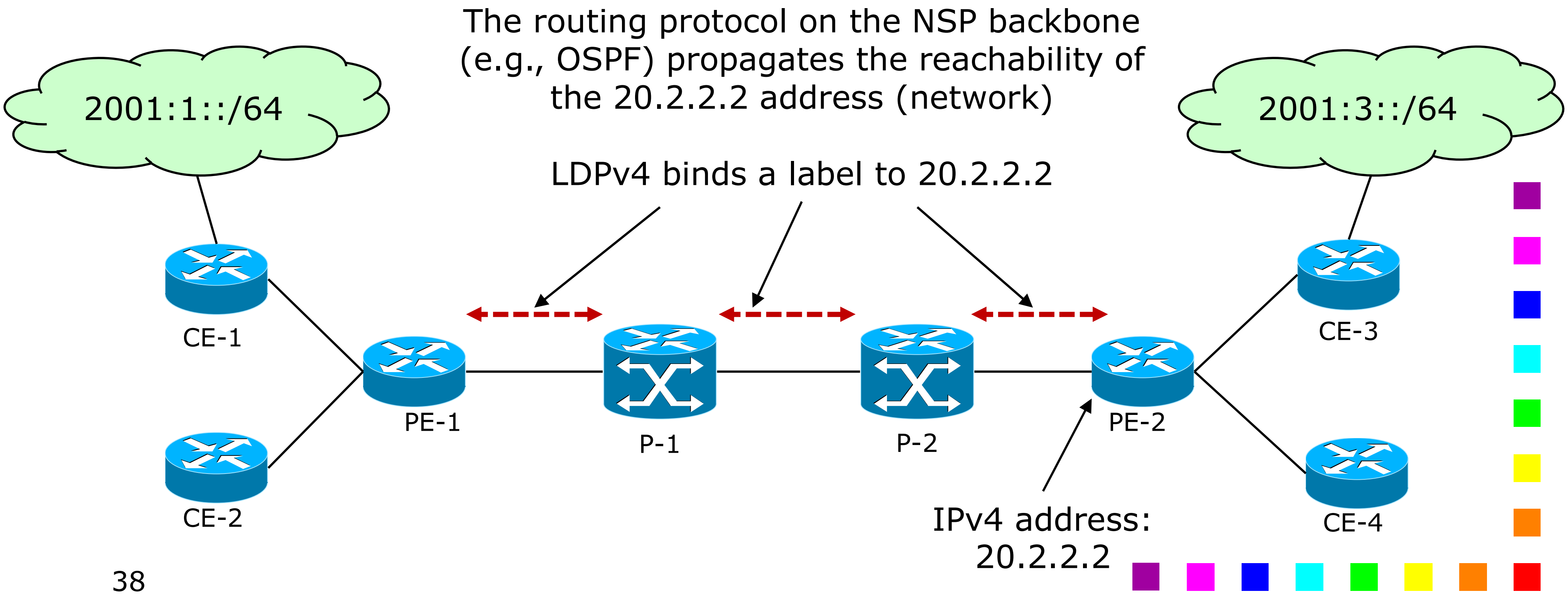
Requirements

- The ISP has to upgrade the Provider Edge (PE) routers to support IPv6 and MP-BGP
- Core routers (P) do not need any change in terms of configuration or software



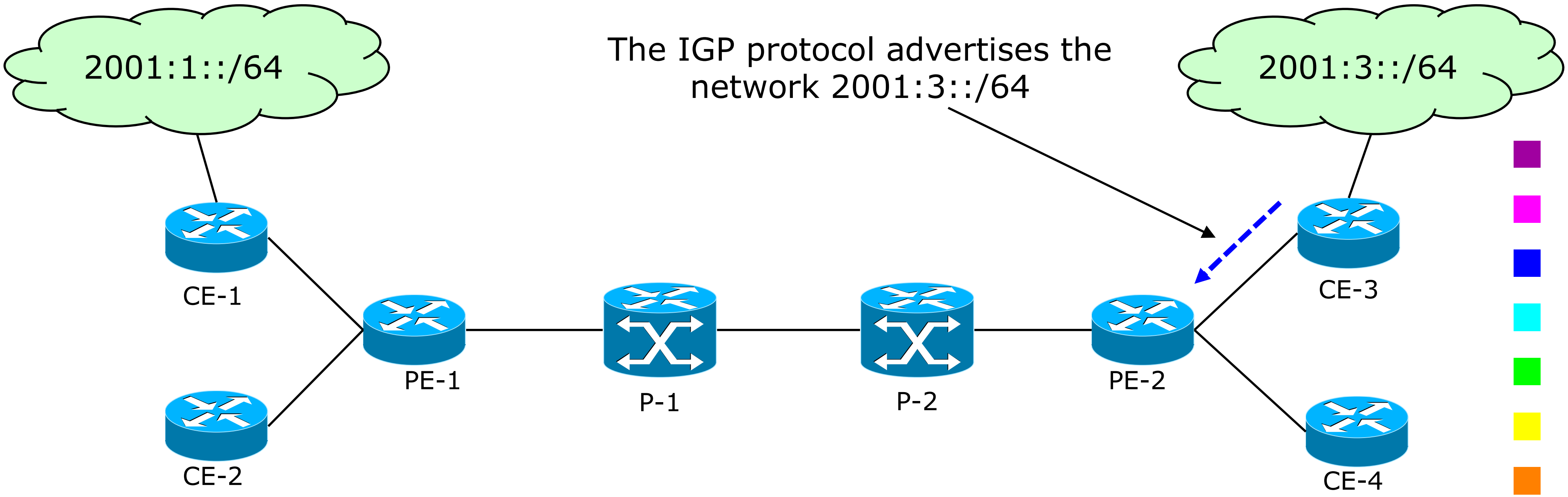
Announcing IPv6 networks (4)

- 6PE routers advertise their IPv4 address (network) into the IGP routing protocol of the IPv4/MPLS core network
- Each router in the MPLS domain will eventually assigns a label corresponding to the route for each 6PE router



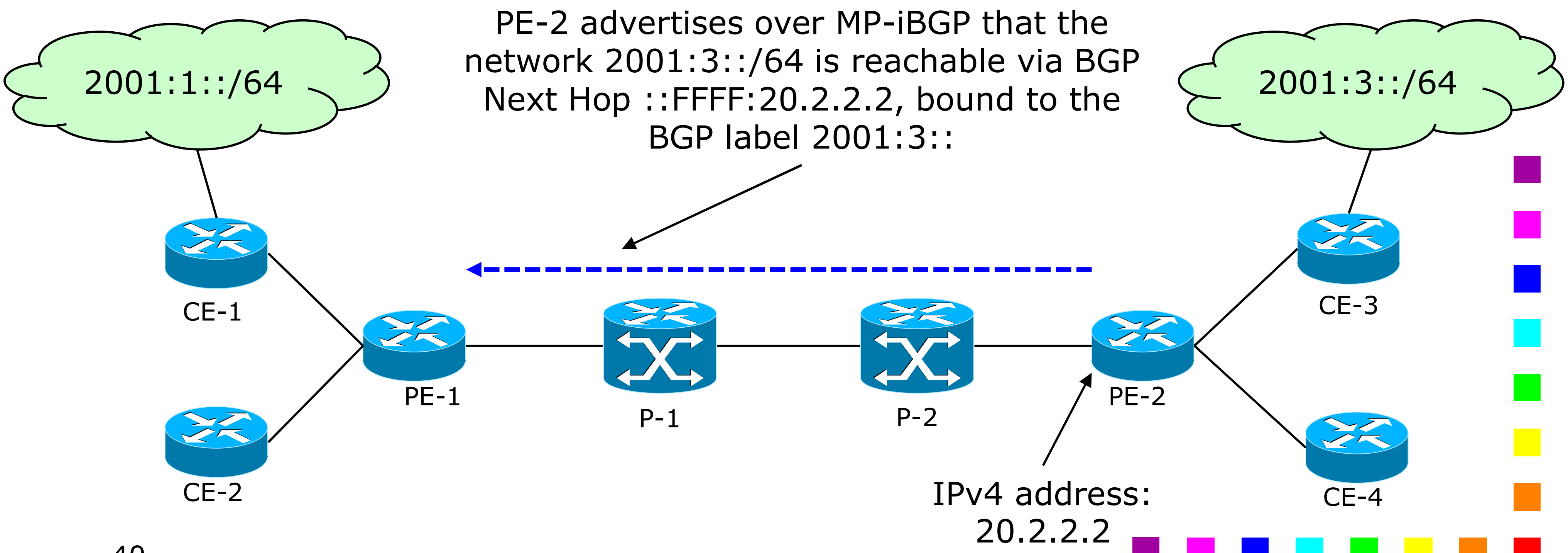
Announcing IPv6 networks (1)

- Customer Edge (CE) routers and 6PE routers are connected with (one or more) logical or physical **native** IPv6 interfaces
- Any common routing protocol (e.g., OSPF, eBGP, static or default routes) between CE and PE allows the distribution of IPv6 reachability information



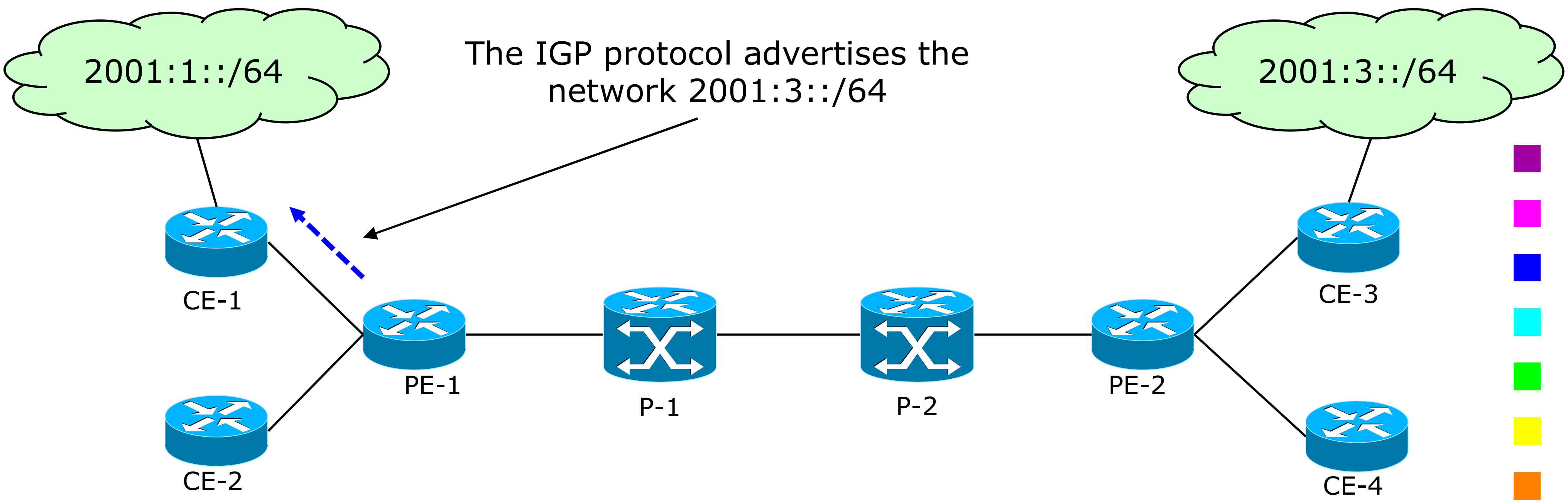
Announcing IPv6 networks (2)

- Customer IPv6 prefixes are exchanged among the 6PE routers over MP-BGP session running over IPv4
- 6PE routers convey their IPv4 address as the BGP Next-Hop for the IPv6 prefixes
  - The BGP Next Hop is the IPv4-mapped IPv6 address of the 6PE



Announcing IPv6 networks (3)

- Finally, the IGP protocol on the other side of the network propagates the advertisement within the network of the other customer
  - This may not needed e.g., in case a default route is used

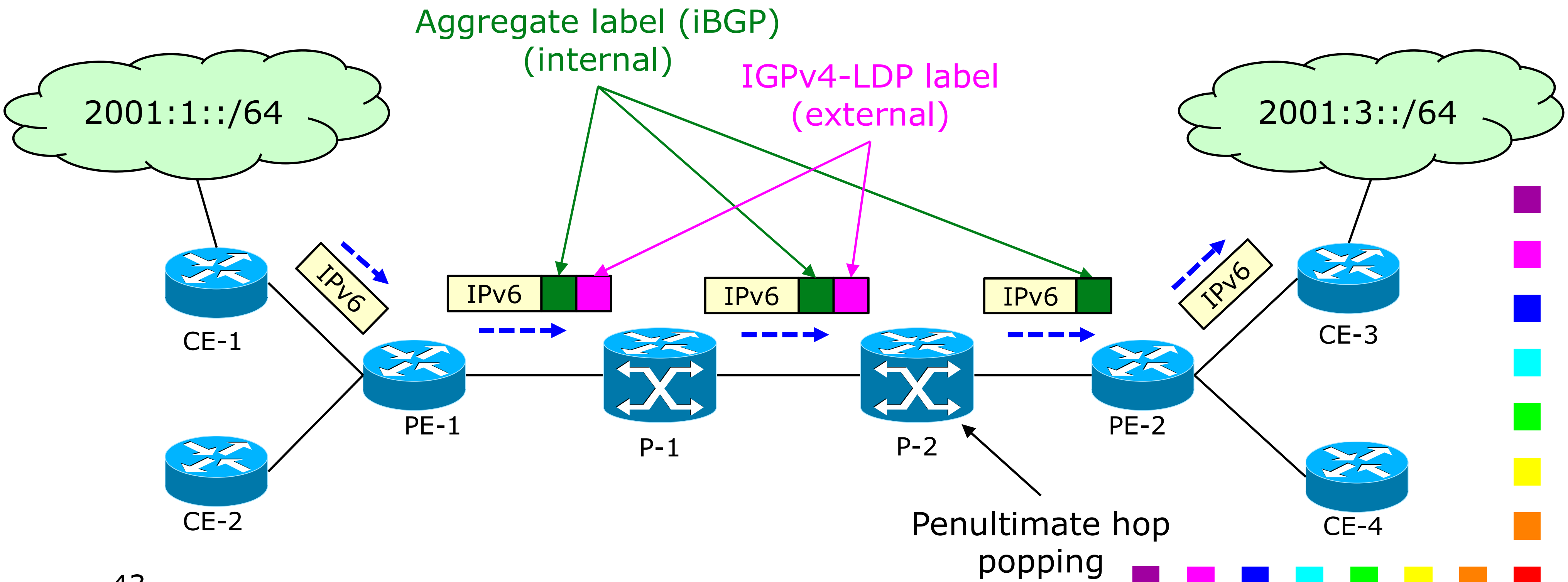


Distributing MPLS labels

- Label distribution and setup is done differently for IPv4 and for IPv6 routes
  - IPv4 labels are distributed through LDPv4 in the core network (P routers)
  - IPv6 labels are distributed with MP-iBGPv4 among the PEs

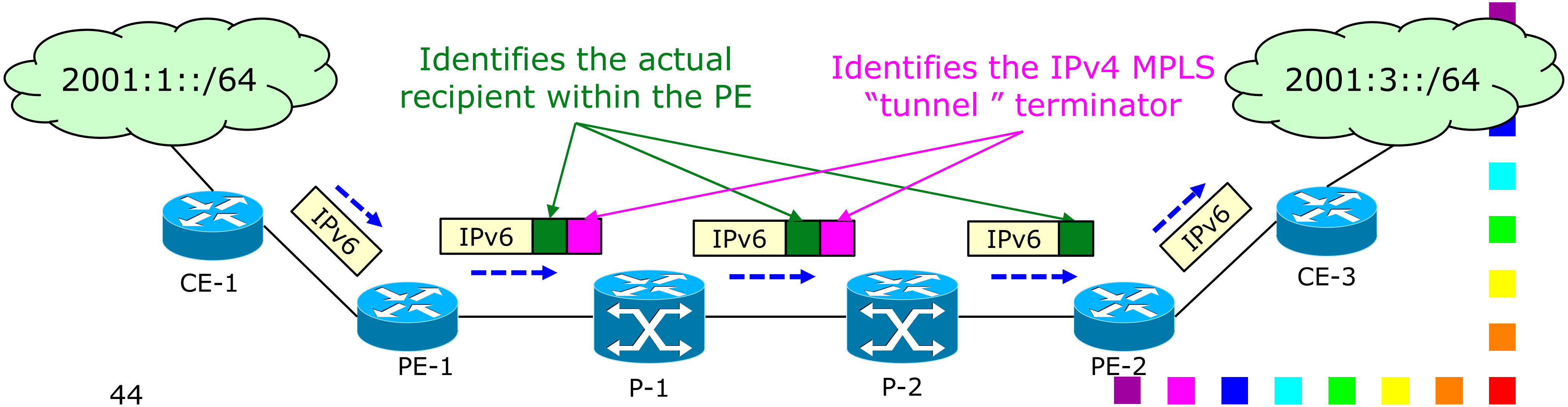
Forwarding IPv6 traffic

- The CE sends a IPv6 packet to PE-1
- The ingress 6PE router tunnels the IPv6 data over an LSP towards a the Egress 6PE router identified by the IPv4 address that derives from the IPv4-mapped IPv6 address of the BGP Next Hop field for the corresponding IPv6 prefix

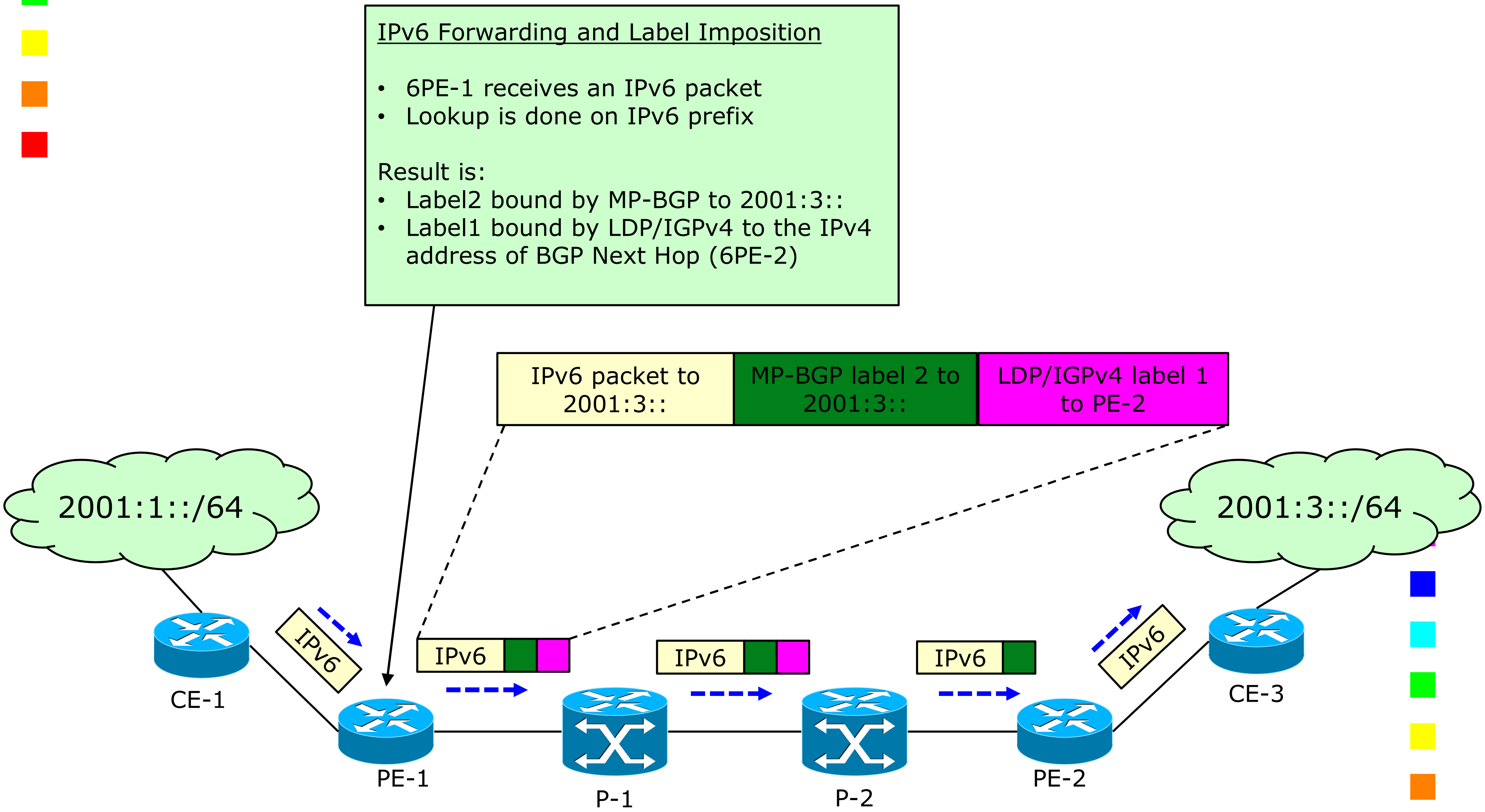


Forwarding IPv6 traffic: inner and outer labels (1)

- 6PE encapsulation involves two labels
  - The inner label ("aggregated IPv6 label") is bounded to each advertised destination IPv6 prefix
  - The outer label is associated with the egress 6PE IPv4 address
- Note that the inner label is in principle not required for the operation, but it keeps the solution the same as for VPNs
  - Without the inner label the "penultimate hop" router would have to be able to forward a plain IPv6 packet to the egress 6PE router



Forwarding IPv6 traffic: inner and outer labels (2)



Conclusions

- PE routers have to be dual stack and to support MP-BGP, while P routers do not need any modification
- Provide native IPv6 services to customers without changing the IPv4 MPLS core network, which turns into minimal operational cost and risk
- 6PE scenario is similar to packet forwarding in MPLS VPNs
  - IPv6 CEs have a single routing peer and do not need any change whenever remote IPv6 CEs are connected or removed
- 6PE fits very well into the general MPLS philosophy
  - However, 6PE does not justify the deployment if MPLS core network
  - Therefore, 6PE is meant for deployment scenarios where the MPLS core is available